

SVKM'S NMIMS

**MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING/
SCHOOL OF TECHNOLOGY MANAGEMENT**

Academic Year: 2024-2025

Program: B.Tech /MBATech/BTI

Year: II/IV Semester: IV/VIII

Stream: Data Science

Subject: Statistical Structures in Data and Inference

Time: 3 hrs (10:00 am to 1:00 pm)

Date: 30/04/2025

No. of Pages: 06

Marks: 100

Final Examination (24-25)/Re Examination (23-24)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No. 1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) **In all 5 questions to be attempted.**
- 4) All questions carry equal marks.
- 5) **Answer to each new question to be started on a fresh page.**
- 6) **Figures in brackets on the right hand side indicate full marks.**
- 7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	[20]
CO-2; SO-1; BL-3	a.	A hospital uses three different brands of COVID-19 test kits—X, Y, and Z. The hospital purchases 30% of its test kits from brand X, 45% from brand Y, and 25% from brand Z. Each brand has different false positive rates: Brand X gives a false positive result in 3% of cases. Brand Y gives a false positive result in 5% of cases. Brand Z gives a false positive result in 2% of cases. A randomly selected test result is found to be a false positive. What is the probability that the test was from brand Y?"	[5]
CO-3; SO-3; BL-2	b.	Explain ANOVA with suitable example and assumptions.	[5]
CO-2; SO-1; BL-3	c.	A pharmaceutical company claims that its new vitamin supplement increases hemoglobin levels by an average of 1.8 g/dL, with a standard deviation of 0.5 g/dL. A health researcher suspects that the actual increase might be different from this claim. To test this, they randomly sample 50 individuals who took the supplement and find that their average hemoglobin increase is 1.6 g/dL. At a 5% significance level, conduct a one-sample Z-test to determine whether the company's claim is statistically valid.	[5]

CO- 3; SO- 6; BL-3	d. A company wants to predict employee productivity (measured in tasks completed per week) based on the number of hours worked per week using a linear regression model. After analyzing the data collected over six months, they observe that : i. The scatter plot of hours worked vs. productivity shows a curve instead of a straight-line trend. ii. The residual plot indicates increasing variance as hours worked increases iii. The residuals are not normally distributed when plotted on a histogram. Based on these observations, identify which regression assumptions are violated and explain the possible consequences of these violations.	[5]
Q2		[20]
CO-3; SO-2; BL-4	a. Using Linear Regression, predict an employee's salary based on their experience, education level, and job position from the dataset given below. Split the dataset into training and testing sets, fit the model, and evaluate its performance. Write the Python code in your answer book and interpret the results. <pre> import pandas as pd import numpy as np # Set random seed for reproducibility np.random.seed(123) n_samples = 200 # Generate synthetic employee salary dataset experience_years = np.random.randint(1, 20, n_samples) # Years of experience education_level = np.random.randint(1, 4, n_samples) # 1: Bachelor's, 2: Master's, 3: PhD job_position = np.random.randint(1, 6, n_samples) # Job level (1-5) # Salary (Target) with some noise salary = (experience_years * 1000) + (education_level * 8000) + (job_position * 3000) + np.random.randn(n_samples) * 10000 # Convert to DataFrame df = pd.DataFrame({ 'Experience (Years)': experience_years, 'Education Level': education_level, </pre>	[10]

		'Job Position': job_position, 'Salary': salary })	
CO-1; SO-1; BL-3	b.	Derive the mathematical expressions for estimating the parameters for a Binomial distribution using Maximum Likelihood Estimation (MLE).	[10]
Q3			[20]
CO-3; SO-2; BL-4	a.	<pre> import numpy as np import pandas as pd import matplotlib.pyplot as plt # Set random seed for reproducibility np.random.seed(42) # Generate synthetic customer data n_samples = 500 age = np.random.normal(loc=35, scale=10, size=n_samples) income = np.random.normal(loc=60000, scale=15000, size=n_samples) web_activity = np.random.uniform(0, 10, size=n_samples) # Generate purchase decision (target variable) based on realistic factors probability = 1 / (1 + np.exp(-(-3 + 0.05 * age + 0.00003 * income + 0.3 * web_activity)))) purchase = (np.random.rand(n_samples) < probability).astype(int) # Create a DataFrame df = pd.DataFrame({ 'Age': age, 'Income': income, 'Web_Activity': web_activity, 'Purchased': purchase }) </pre> A company wants to predict whether a customer will purchase a product based on their age, income, and online activity score. Given a dataset with these features, predict the customer will purchase the product both for training and testing data set based on their	[10]

		age, fare, and class. Write the Python code in your answer book and provide an interpretation of the results.	
CO-2; SO-1; BL-3	b.	How can we estimate the parameters mean and standard deviation of a normal distribution using Maximum Likelihood Estimation (MLE) in the below data set by implementing through python code. <pre>import numpy as np import scipy.optimize as opt import scipy.stats as stats # Generate example data from a normal distribution np.random.seed(42) data = np.random.normal(loc=10, scale=3, size=100)</pre>	[10]
Q4			[20]
CO-2; SO-6; BL-3	a.	A researcher is analyzing the effectiveness of two different training programs on employee performance. Two independent groups of employees were trained using different methods, and their post-training scores were recorded. The first group of 50 employees had an average performance score of 75 with a standard deviation of 8, while the second group of 65 employees had an average score of 70 with a standard deviation of 10. (a) Compute the estimated standard error of the difference between the two means. (b) At a significance level of $\alpha = 0.05$, test whether there is a significant difference in the mean performance scores between the two groups. Write a Python code and interpret the results.	[10]
CO-2; SO-6; BL-3	b.	A university wants to evaluate the effectiveness of an online learning platform by comparing student performance before and after using the platform. A random sample of 12 students took a test before using the platform and then took a similar test after one month of studying with the online resources. The scores (out of 100) are recorded as follows: Before: 55, 62, 70, 58, 65, 74, 80, 69, 72, 60, 77, 85 After: 65, 70, 75, 63, 72, 79, 85, 74, 78, 68, 82, 90	[10]

		Perform a paired t-test to determine whether the online learning platform has significantly improved student performance. • Write a Python program to conduct the test. • Interpret the results and conclude whether the learning platform had a significant impact.																																																	
Q5 CO-3; SO-2; BL-4		Define Lasso regression with their uses and application. Using Lasso regression, predict the body mass of penguins based on their bill_length_mm, bill_depth_mm, flipper_length_mm, island, sex and species. Split the dataset into training and testing sets, train the model, and evaluate its performance. Write the Python code in your answer book and interpret the results. <pre>import pandas as pd import seaborn as sns # Load dataset and drop missing values df = sns.load_dataset("penguins").dropna()</pre>	[20]																																																
Q 6			[20]																																																
CO2-; SO-1; BL-3	a.	A large hospital hires most of its nurses from the two major universities in the area. Over the last year, they have been giving a test to the newly graduated nurses entering the hospital to determine which school, if either, seems to educate its nurses better. Based on the following scores (out of 100 possible points), help the personnel office of the determine whether the schools differ in quality. Use the Mann whitney U test with a 10% level of significance. <table><tr><td></td><td colspan="7">Test Score</td></tr><tr><td>School A</td><td>97</td><td>69</td><td>73</td><td>84</td><td>76</td><td>92</td><td>90</td></tr><tr><td>School B</td><td>88</td><td>99</td><td>65</td><td>69</td><td>97</td><td>84</td><td>85</td></tr><tr><td></td><td colspan="7">Test Score</td></tr><tr><td>School A</td><td>88</td><td>84</td><td>87</td><td>93</td><td></td><td></td><td></td></tr><tr><td>School B</td><td>89</td><td>91</td><td>90</td><td>87</td><td>91</td><td>72</td><td></td></tr></table>		Test Score							School A	97	69	73	84	76	92	90	School B	88	99	65	69	97	84	85		Test Score							School A	88	84	87	93				School B	89	91	90	87	91	72		[10]
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CO-3; SO-2; BL-4	b.	Bank of Lincoln is interested in reducing the amount of time people spend waiting to see a personal banker. The bank is interested in the relationship between waiting time (Y) in minutes and number of bankers on duty (X). Customers were randomly selected with the data given below: <table><tr><td>X</td><td>2</td><td>3</td><td>5</td><td>4</td><td>2</td><td>6</td><td>1</td><td>3</td><td>4</td><td>3</td><td>3</td><td>2</td><td>4</td></tr><tr><td>Y</td><td>12.8</td><td>11.3</td><td>3.2</td><td>6.4</td><td>11.6</td><td>3.2</td><td>8.7</td><td>10.5</td><td>8.2</td><td>11.3</td><td>9.4</td><td>12.8</td><td>8.2</td></tr></table> Calculate mathematically the regression equation that best fit the data.	X	2	3	5	4	2	6	1	3	4	3	3	2	4	Y	12.8	11.3	3.2	6.4	11.6	3.2	8.7	10.5	8.2	11.3	9.4	12.8	8.2	[10]
X	2	3	5	4	2	6	1	3	4	3	3	2	4																		
Y	12.8	11.3	3.2	6.4	11.6	3.2	8.7	10.5	8.2	11.3	9.4	12.8	8.2																		
Q 7			[20]																												
CO-2; SO-1; BL-3	a.	MacroSwift estimated last year that 35% of potential software buyers planned to wait for an upgrade before purchasing the new operating system, Window Panes. After an advertising campaign, a survey of 3,000 people found 950 were still skeptical. At the 5% significance level, can the company conclude that the proportion of skeptical people has decreased? Write the Python code in your answer book and interpret the results.	[10]																												
CO-3; SO-2; BL-4	b.	Using the "tips" dataset from Seaborn, analyze whether the total bill and tip amounts are significantly affected by the group size and the day of the week.Using Multivariate Analysis of Variance (MANOVA), test whether there is a statistically significant difference in the total bill and tip amount based on these factors. <pre>import pandas as pd import seaborn as sns import numpy as np # Load the tips dataset df = sns.load_dataset("tips").dropna()</pre>	[10]																												